Environmental impacts

Challenge 2

LLM marked question

30 attempts remaining today

The rapid advancements in mobile phone technology have contributed to a culture of frequent upgrades, where consumers are encouraged to buy new phones regularly. While this trend offers numerous benefits, it also comes with significant environmental consequences. Understanding both the positive and negative impacts of this upgrade culture on the environment is critical to fostering responsible consumer behaviour.

Give one negative and one positive environmental impact of the mobile phone upgrade culture.

[2 marks]

Quiz:
STEM SMART Computer Science Week 43

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Long answer - autonomous machines

Part A Evaluate the answer

Question: A company is developing software for a fleet of self-driving cars to provide pre-bookable transport services in a large city. Discuss the moral and legal issues that the software company will need to consider.

A student has written the following answer. Mark each part of the answer with whether the sentence is:

- **P** - A new point being made
- **E** - Evidence or an extension of the point being made
- **X** - Incorrect information, or does not add anything new

There are lots of moral and legal issues around self-driving cars and this is a topic that has been in the news recently ( ). If a self-driving car ends up in a dangerous situation where an accident is inevitable, the programmers must decide in advance whether passenger or pedestrian safety is a higher priority ( ). It is also unclear who would be responsible if a self-driving car had an accident ( ).

If someone needs to get home late at night, the self-driving car's location, and footage from internal cameras, may be able to be tracked by family waiting at home ( ). This may make some people feel safer compared to travelling in a taxi with a human driver, but some people may find this an infringement of privacy, especially if it is not possible to turn this feature off ( ).

Items:

P E X

Part B How many marks?

If this was the mark scheme, decide how many marks you would award for this student's answer. Pick the most appropriate mark band.

Marks	Description
5	Three or more relevant points have been made, and the answer covers both legal and moral issues. There is sufficient detail to demonstrate logical reasoning and a good level of understanding.
3–4	The answer covers both legal and moral issues. The answer is logically structured and shows reasonable understanding, but lacks detail in some areas.
1–2	A few relevant points are made but they may not relate specifically to the scenario. Some understanding of the issues is shown.

- ○ 5
- ○ 3–4
- ○ 1–2

Quiz:
STEM SMART Computer Science Week 43

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Long answer - algorithmic bias

Part A Evaluate the answer

Question: A bank is considering purchasing automated decision-making software so that it can quickly decide whether a customer can take out a loan. Discuss the issues the bank would have to consider before rolling out this software.

A student has written the following answer. Mark each part of the answer with whether the sentence is:

- **P** - A new point being made
- **E** - Evidence or an extension of the point being made
- **X** - Incorrect information, or does not add anything new

Using automated decision-making software is not a good idea for the bank ( ). The software may be biased against certain groups of people ( ). If someone had bad credit, they might be refused a loan and that would mean the software was biased ( ). It also depends how the software was trained because if the data it was trained on was no good, this would not give good results for the customers ( ). However, on the positive side, the bank might be able to make decisions more quickly using this software ( ).

Items:

P E X

Part B How could they improve?

The mark scheme for this question is below. The answer the student has given would receive **1–2 marks**.

Marks	Description
5	Three or more relevant points have been made. There is sufficient detail to demonstrate logical reasoning and a good level of understanding.
3–4	The answer is logically structured and shows a reasonable understanding, but lacks detail.
1–2	A few different points are made but they may not relate specifically to the scenario. Some understanding of the issues is shown. There is little evidence of logical reasoning.

Select **two** things the student could do to improve their answer.

- [ ] Add reasoned consequences for the points they have already made
- [ ] Write more so that the answer is at least three paragraphs
- [ ] Remove some information that is incorrect from the answer
- [ ] Make some additional points

Quiz:
STEM SMART Computer Science Week 43

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Long answer - remote working

Challenge 2

Part A Evaluate the answer

Question: Since the COVID-19 pandemic, many employers are asking employees who were previously working remotely to return to working in the office. Discuss the ethical responsibilities the employer should consider when making a change to their remote working policies.

A student has written the following answer. Mark each part of the answer with whether the sentence is:

- **P** - A new point being made
- **E** - Evidence or an extension of the point being made
- **X** - Incorrect information, or does not add anything new

An employer may want employees to return to working in the office because the costs associated with running an office such as rent, heating, and lighting are expensive for the business, and do not decrease by much if no one is working there ( ).

Asking employees to work more frequently at the office has an impact on the employee's daily life, for example, if the employee now needs to commute to work, this will take time and be more expensive than working from home ( ). The employer must be careful that any changes in their policy do not have a disproportionate impact on a particular group of employees, for example, employees with young children might find that they are no longer able to drop off their children at school and get to work on time ( ).

The employer must also consider whether their working practices are inclusive of all employees regardless of whether they are working remotely or at home ( ). For example, if key information is written on a whiteboard in the office, this will not be visible to employees working remotely, and might result in those employees feeling isolated and unable to fully participate ( ).

Items:

P E X

Part B How could they improve?

The mark scheme for this question is below. The answer the student has given would receive **3–4 marks**.

Marks	Description
5	Three or more relevant points have been made. There is sufficient detail to demonstrate logical reasoning and a good level of understanding.
3–4	The answer is logically structured and shows a reasonable understanding, but lacks detail.
1–2	A few different points are made but they may not relate specifically to the scenario. Some understanding of the issues is shown. There is little evidence of logical reasoning.

What should the student do to improve their answer so that it receives 5 marks?

- ○ Make some additional points
- ○ Add reasoned consequences for the points they have already made
- ○ Write more so that the answer is at least three paragraphs
- ○ Remove some information that is incorrect from the answer

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Artificial intelligence and bias

LLM marked question

A university is developing an AI system to assist with undergraduate admissions decisions.

The system is trained on ten years of historical admissions data, including applicants' predicted grades, personal statements, school attended, postcode, social background, disability status, and prior admission outcomes.

The AI will generate a suitability score for each applicant and recommend whether they should receive an offer, be placed on a waiting list, or be rejected.

Due to increasing application numbers and limited staff resources, the university intends to rely heavily on the AI's recommendations, with only a small percentage of applications being reviewed manually.

Discuss the ethical issues involved in the use of AI in the university admissions process.

In your answer you should consider impacts on individuals, society and the university.

Part A Data collection

30 attempts remaining today

Explain two ethical concerns related to the collection of data used to train the AI system. [2 marks]

[2 marks]

Part B Bias and fairness

30 attempts remaining today

Explain how bias in training data could lead to unfair outcomes from the AI system [2 marks]

[2 marks]

Part C Transparency and accountability

30 attempts remaining today

Explain why transparency and accountability are important when using AI in the admissions process. [2 marks]

[2 marks]

Part D Social impact

30 attempts remaining today

Explain one positive and one negative social impact of using AI in the university's admissions process [2 marks]

[2 marks]

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Encryption

LLM marked question

A company is developing a cloud-based system.

Users log in with a username and password which is stored in the database. Users can also upload confidential documents, which are transmitted over the internet between their device and the server.

The company wants to protect:

- Stored passwords if the database is accessed without authorisation.
- Documents while they are being transmitted.

Security measures must not significantly reduce the system performance.

Compare the use of encryption and hashing in this system.

Part A

30 attempts remaining today

Describe and compare how encryption and hashing work

[3 marks]

Part B

30 attempts remaining today

Explain how hashing could be used in this system and why this would improve security.

[2 marks]

Part C

30 attempts remaining today

Explain how encryption could be used in this system and why this would improve security

[2 marks]

Part D

30 attempts remaining today

The company is concerned that security measures must not significantly reduce system performance.

Evaluate whether the security benefits of encryption and hashing outweigh their impact on system performance in this scenario.

[3 marks]

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